INCH-POUND

MIL-M-38510/336A
AMENDMENT 4
26 June 1993
SUPERSEDING
AMENDMENT 3
27 October 1987

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL,
SHIFT REGISTERS, CASCADABLE, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/336A, dated 9 August 1983 and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.1, under circuit: Add "cascadable" after register.

1.2.3, under letter: Delete "X" and substitute "2".

1.2.3, case outline, add: "X C2-A (20-terminal, .350" x .350"), square chip carrier package)".

1.3, thermal resistance: Delete in its entirety (as substituted in amendment 2) and substitute the following:

"Thermal resistance, junction-to-case (Θ_{JC}):
Cases E, F, 2, and X - - - - - - - - - - - - - - - (See MIL-M-38510, appendix C)".

1.3, after Junction temperature (T_J): Add "2/".

At bottom of page, delete footnotes 2/ and 3/ (as added in amendment 2) and substitute the following:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening condition in accordance with method 5004 of MIL-STD-883."

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

	Replacement page	Page replaced
	9	9
	10	10
	13	13
*	14	14

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.

PAGE 2

1.4, after supply voltage: Add "V_{CC}".

1.4: Add "Width of clock pulse low, $t_{PW(CPL)}$ - - - - - - - - 6.0 ns minimum".

1.4, setup time data high to clock pulse, $t_{SD(H)}$: Delete "5.0 ns" and substitute "6.0 ns".

1.4, setup time select high to clock pulse, $t_{SS(H)}$: Delete "10.0 ns" and substitute "10.5 ns".

PAGE 3

TABLE I, add the following:

"

Maximum frequency	f_{MAX}	V_{CC} = 5.0 V C_L = 50 pF ±10% See figure 5	All	80	---	MHz

"

TABLE I, footnote 1/, delete and substitute:

"1/ Not more than one output should be shorted at a time."

PAGE 4

3.2.3, delete and substitute:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request."

3.6, delete and substitute:

"3.6 Marking. Marking shall be in accordance with MIL-M-38510."

TABLE II, interim electrical parameters: Delete "(pre burn-in)".

TABLE II, group A test requirements, under class B devices: Add "8".

PAGE 5

4.2a(1): Delete "D or E" and substitute "D, E, or F".

4.2c, delete and substitute:

"c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510."

PAGE 6

FIGURE 1: Delete "Case X" and substitute "Cases 2 and X".

PAGE 12

TABLE III: Delete "Case X" and substitute "Cases 2 and X".

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.

PAGE 15

4.4.3c(1): Delete "D or E" and substitute "D, E, or F".

4.4.3c(3), delete and substitute:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

6.1: Delete in its entirety.

PAGE 16

TABLE IV: Add "X" to circuit C column.

The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no lialbility whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

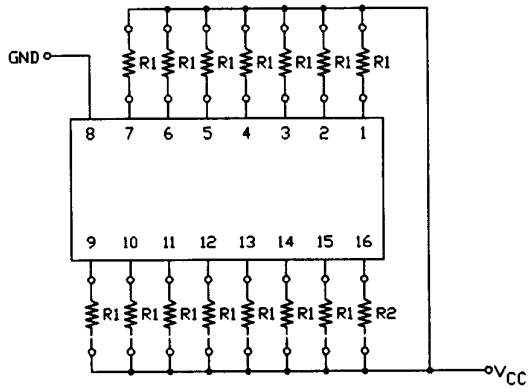
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1345)

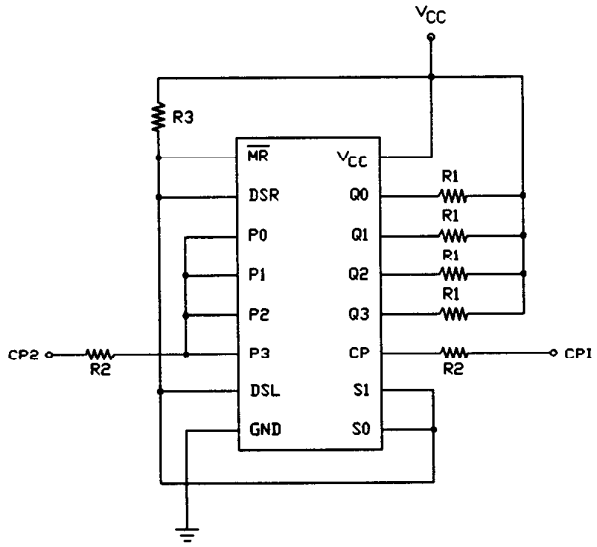
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.

Device type 01 excluding cases 2 and X
High temperature (175°C) burn-in (optional circuit)



NOTES:
1. R1 = 2 kΩ ±5%, R2 = 2Ω.
2. V_{CC} shall be high enough to ensure 5.0 V at the device terminal.

Device type 01
Cases E, F, 2, and X
(Pins not referenced for cases 2 and X are N/C)

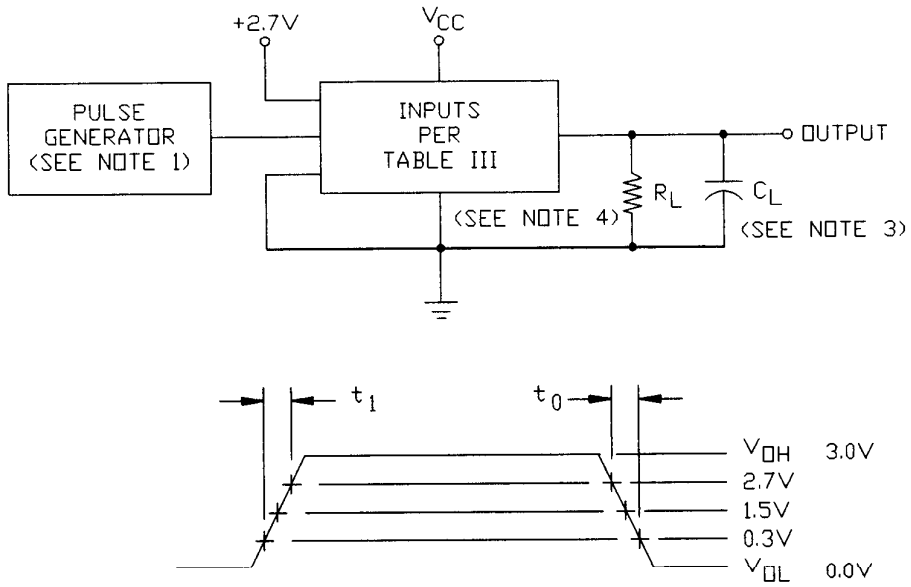


NOTES:
1. CP = 100 kHz ±50% square wave, duty cycle = 50 ±15%, CP2 = CP1 ÷ 2, V_{IL} = -0.5 V minimum to 0.8 V maximum, V_{IH} = 2.0 V minimum to 5.5 V maximum.
2. R1 = 240Ω ±5%, R2 = 51Ω ±5%, R3 = 1 kΩ ±5%.
3. V_{CC} and R4 shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

Supersedes page 9 of MIL-M-38510/336A of 9 August 1983.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.



NOTES:
1. Pulse generator has the following characteristics: $t_1 = t_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} \approx 50\Omega$.
2. Inputs not under test are at ground.
3. C_L = 50 pF ±10% including scope probe, wiring and stray capacitance without package in test fixture.
4. $R_L = 499\Omega$ ±5%.
5. Voltage measurements are to be made with respect to network ground terminal.

FIGURE 5. Switching time waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2,X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	DSR	P0	P1	P2	P3	DSL	GND	SO	S1	CP	Q3	Q2	Q1	Q0	V_{CC}				
1 T_C = 25°C	I_{OS}	3011	53	5.5 V		5.5 V	0 V	0 V	0 V		GND	5.5 V	5.5 V	3/				0 V	5.5 V	Q0	-60	-150	mA
		"	54	"		0 V	5.5 V	5.5 V	"		"	"	"	"	0 V				"	Q1	"	"	"
		"	55	"		"	0 V	0 V	5.5 V		"	"	"	"		0 V			"	Q2	"	"	"
		"	56	"		"	"	"	"		"	"	"	"			0 V		"	Q3	"	"	"
	I_{OD}		57	0 V							"				2.5 V				4.5 V	Q3	60		"
			58	"							"					2.5 V			"	Q2	"		"
			59	"							"						2.5 V		"	Q1	"		"
			60	"							"							2.5 V	"	Q0	"		"
	I_{CC}	3005	61	5.5 V	5.5 V	0 V	0 V	0 V	0 V	5.5 V	"	5.5 V	5.5 V	3/					5.5 V	V_{CC}		46	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 T_C = 25°C	Functional test 5/	3014	62-110																6/	All outputs			

See footnotes at end of table

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T14:17Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥ 2.0 V, low ≤ 0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases E,F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2,X 1/	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20		Min	Max	
			Test no.	$\overline{MR}$	DSR	P0	P1	P2	P3	DSL	GND	S0	S1	CP	Q3	Q2	Q1	Q0	V_{CC}				
8	Repeat subgroup 7 tests at T_C = +125°C and T_C = -55°C.																						
9 T_C = 25°C	t_{PLH1}	3003	111	2.7 V		2.7 V	2.7 V	2.7 V	2.7 V		GND	2.7 V	2.7 V	IN	OUT	OUT	OUT	OUT	OUT	5.0 V	CP to Q0	3.5	7.0	ns
		"	112	"							"	"	"	"						"	CP to Q1	"	"	"
		"	113	"							"	"	"	"						"	CP to Q2	"	"	"
		"	114	"							"	"	"	"						"	CP to Q3	"	"	"
	t_{PHL1}	3003	115	"		0 V	0 V	0 V	0 V		"	"	"	"	OUT	OUT	OUT	OUT	OUT	"	CP to Q0	"	"	"
		"	116	"							"	"	"	"						"	CP to Q1	"	"	"
		"	117	"							"	"	"	"						"	CP to Q2	"	"	"
		"	118	"							"	"	"	"						"	CP to Q3	"	"	"
	t_{PHL2}	3003	119	IN		2.7 V	2.7 V	2.7 V	2.7 V		"	"	"		OUT	OUT	OUT	OUT	OUT	"	$\overline{MR}$ to Q0	4.5	12.0	"
		"	120	"							"	"	"							"	$\overline{MR}$ to Q1	"	"	"
		"	121	"							"	"	"							"	$\overline{MR}$ to Q2	"	"	"
		"	122	"							"	"	"							"	MR to Q3	"	"	"
	f_{MAX} 7/	3003	123	2.7 V	IN	IN	IN	IN	IN	IN	"	2.7 V	0.0 V	IN	OUT	OUT	OUT	OUT	OUT	"	Q0	100		MHz
		"	124	"							"	0 V	2.7 V	"						"	Q3	"		"
		"	125	"							"	2.7 V	"	"						"	Q0	"		"
		"	126	"							"	"	"	"						"	Q1	"		"
		"	127	"							"	"	"	"						"	Q2	"		"
		"	128	"							"	"	"	"						"	Q3	"		"
10	t_{PLH1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																				3.0	8.5	ns
	t_{PHL1}																					3.0	8.5	ns
	t_{PHL2}																					4.5	14.5	ns
	f_{MAX} 7/																					80		MHz
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																							

1/ For cases 2 and X pins not referenced are N/C.
2/ Apply 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to $\overline{MR}$ then 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to CP prior to making measurement.
3/ Apply 3.0 V min/5.5 V max, 0 V, 3.0 V min/5.5 V max to CP prior to making measurement.
4/ I_{IL} limits shall be as follows:

Parameter	Test no.	Min/Max limits in mA for circuit		
		A	B	C
I_{IL1}	43 through 52	-.25/-.60	-.03/-.60	-.03/-.60

5/ A = 2.5 V minimum, B = 0.5 V maximum, X = don't care.
H ≥ 1.5 V, L ≤ 1.5 V.
6/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.
7/ f_{MAX} shall be measured only under the conditions of initial qualification and after process or design changes which may affect this parameter. For all other conditions, f_{MAX} shall be guaranteed, if not tested, to the limits specified in table III, herein.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T14:17Z
Check the source to verify that this is the current version before use.